



TED UNIVERSITY
Department of Computer Engineering

ADA 442 - PROJECT

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1. Introduction

This project focuses on a binary classification problem using the Bank Marketing dataset. The aim is to predict whether a bank client will subscribe to a term deposit based on customer information, campaign details, and economic indicators. The project follows a complete machine learning workflow that includes data cleaning, preprocessing, feature engineering, feature selection, model comparison, hyperparameter tuning, evaluation, and deployment with Streamlit.

2. Dataset and Data Cleaning

The dataset contains 4,119 observations and 21 original columns. It includes numerical variables such as age, campaign count, previous contacts, employment variation rate, consumer price index, Euribor rate, and number of employees. It also includes categorical variables such as job, education, housing loan, personal loan, contact type, month, and previous campaign outcome. The target variable is y, where yes represents subscription and no represents no subscription.

Initial checks were performed to examine dataset shape, column types, duplicate rows, missing values, and target distribution. No regular missing values were found. Categorical values such as unknown were treated as valid categories because they can still carry information about unavailable customer details. The duration column was removed before modeling because call duration is only known after a phone call is completed. Using this column would create data leakage and make the model unrealistic for real prediction use.

3. Preprocessing and Feature Engineering

The target variable was converted from yes and no into 1 and 0. Numerical and categorical features were separated before modeling. Numerical variables were scaled using StandardScaler, while categorical variables were transformed using OneHotEncoder with unknown categories safely ignored. These steps were combined inside a ColumnTransformer so that preprocessing remains consistent and repeatable.

Several engineered features were added to improve the information available to the model. Examples include has_university_degree, is_high_campaign, white_collar, blue_collar, other_collar, previous_contact, has_multiple_loans, economic_stress, econ_interact, age_euribor_interact, campaign_conf_interact, and cpi_euribor_diff. Feature selection was applied with SelectKBest and mutual information to keep the most relevant encoded features and reduce unnecessary noise.

4. Model Training and Evaluation

The dataset was split into training and test sets using stratification so that the target class ratio stayed similar in both sets. Since the target classes are imbalanced, SMOTE was included in the modeling pipeline to increase the representation of the minority class during training. Three classification models were compared: Logistic Regression, Random Forest Classifier, and Gradient Boosting Classifier.

The models were evaluated using accuracy, precision, recall, F1-score, ROC-AUC, confusion matrix, and classification report. F1-score and recall were especially important because the positive class is much smaller than the negative class. Random Forest was selected as the final model because it provided a strong balance between overall performance and minority class detection. GridSearchCV was used to tune the Random Forest model, including parameters such as number of estimators, maximum depth, minimum samples split, and the number of selected features.

Final pipeline structure: preprocessing -> feature selection -> SMOTE -> tuned Random Forest Classifier -> saved model file rf_pipeline.sav

5. Deployment and Recommendations

The final model was deployed through a Streamlit web application named Bank Campaign Subscription Predictor. The app contains two main tabs: an Exploratory Data Analysis tab and a Customer Prediction tab. The EDA tab displays target distribution, duration leakage information, numerical correlation heatmap, and a dataset preview. The prediction tab allows a user to enter customer and campaign information and receive a yes or no prediction with probability.

Streamlit Cloud link: <https://i4mddnuxhdgtcwutbpmdjm.streamlit.app/>

For future improvement, the project can be extended by testing more models, collecting more recent campaign data, monitoring model performance after deployment, and improving the interface with clearer explanations for business users. The current system already provides an end-to-end machine learning workflow, from raw data analysis to a usable prediction interface.